

```

1 V = 5
2 edges = [(0,1,-1),(0,2,4),(1,2,3),(1,3,2),(1,4,2),(3,2,5),(3,1,1),(4,3,-3)]
3
4 dist = [999, 999, 999, 999, 999]
5 dist[0] = 0
6
7 for i in range(4):
8     for u, v, w in edges:
9         if dist[u] + w < dist[v]:
10             dist[v] = dist[u] + w
11
12         print("Vertex Distance")
13
14         for i in range(5):
15             print(i, dist[i])

```

Vertex Distance

```

0 0
1 -1
2 999
3 999
4 999

```

Vertex Distance

```

0 0
1 -1
2 4
3 999
4 999

```

Vertex Distance

```

0 0
1 -1
2 2

```

```

1 INF = 999
2
3 a = [
4     [0, 5, INF, 10],
5     [INF, 0, 3, INF],
6     [INF, INF, 0, 1],
7     [INF, INF, INF, 0]
8 ]
9
10 n = 4
11
12 for k in range(n):
13     for i in range(n):
14         for j in range(n):
15             if a[i][k] + a[k][j] < a[i][j]:
16                 a[i][j] = a[i][k] + a[k][j]
17
18 for row in a:
19     print(row)

```

input

```

[0, 5, 8, 9]
[999, 0, 3, 4]
[999, 999, 0, 1]
[999, 999, 999, 0]

```

```
1 a = [  
2     [0, 1, 0, 0],  
3     [0, 0, 1, 0],  
4     [0, 0, 0, 1],  
5     [0, 0, 0, 0]  
6 ]  
7  
8 n = 4  
9  
10 for k in range(n):  
11     for i in range(n):  
12         for j in range(n):  
13             if a[i][k] == 1 and a[k][j] == 1:  
14                 a[i][j] = 1  
15  
16 for row in a:  
17     print(row)
```



```
[0, 1, 1, 1]  
[0, 0, 1, 1]  
[0, 0, 0, 1]  
[0, 0, 0, 0]
```

```

15 for i in range(n):
16     d[i][i] = 0
17
18 for u, v, w in edges:
19     d[u][v] = w
20     d[v][u] = w
21
22 for k in range(n):
23     for i in range(n):
24         for j in range(n):
25             if d[i][k] + d[k][j] < d[i][j]:
26                 d[i][j] = d[i][k] + d[k][j]
27
28 answer = 0
29 minimum = n
30
31 for i in range(n):
32     count = 0
33
34     for j in range(n):
35         if i != j and d[i][j] <= threshold:
36             count = count + 1
37
38     if count <= minimum:
39         minimum = count
40         answer = i
41
42 print(answer)

```



```

1 times = [
2     [2, 1, 1],
3     [2, 3, 1],
4     [3, 4, 1]
5 ]
6
7 n = 4
8 source = 2
9
10 d = [999] * (n + 1)
11 d[source] = 0
12
13 for i in range(n - 1):
14     for u, v, w in times:
15         if d[u] != 999 and d[u] + w < d[v]:
16             d[v] = d[u] + w
17
18 answer = max(d[1:])
19
20 if answer == 999:
21     print(-1)
22 else:
23     print(answer)

```



```
1 n = 4
2
3 flights = [
4     [0, 1, 100],
5     [1, 2, 100],
6     [2, 3, 100],
7     [0, 3, 500]
8 ]
9
10 src = 0
11 dst = 3
12 K = 2
13
14 d = [999] * n
15 d[src] = 0
16
17 for i in range(K + 1):
18     temp = d[:]
19
20     for u, v, cost in flights:
21         if d[u] != 999 and d[u] + cost < temp[v]:
22             temp[v] = d[u] + cost
23
24     d = temp
25
26 print(d[dst])
```

```

1 V = 4
2
3 edges = [
4     (0, 1, 1),
5     (1, 2, -1),
6     (2, 3, -1),
7     (3, 0, -1)
8 ]
9
10 d = [0] * V
11
12 for i in range(V - 1):
13     for u, v, w in edges:
14         if d[u] + w < d[v]:
15             d[v] = d[u] + w
16
17 found = False
18
19 for u, v, w in edges:
20     if d[u] + w < d[v]:
21         found = True
22         break
23
24 if found:
25     print("Negative Weight Cycle Exists")
26 else:
27     print("No Negative Weight Cycle")

```



Negative Weight Cycle Exists


```
1 a = [  
2     [0, 1, 0],  
3     [0, 0, 1],  
4     [0, 0, 0]  
5 ]  
6  
7 u = 0  
8 v = 2  
9 n = 3  
10  
11 for k in range(n):  
12     for i in range(n):  
13         for j in range(n):  
14             if a[i][k] == 1 and a[k][j] == 1:  
15                 a[i][j] = 1  
16  
17 if a[u][v] == 1:  
18     print("Path Exists")  
19 else:  
20     print("Path Does Not Exist")
```



Path Exists


```
1 a = [  
2     [0, 1, 0, 0],  
3     [0, 0, 1, 0],  
4     [0, 0, 0, 1],  
5     [0, 0, 0, 0]  
6 ]  
7  
8 source = 0  
9 n = 4  
10  
11 for k in range(n):  
12     for i in range(n):  
13         for j in range(n):  
14             if a[i][k] == 1 and a[k][j] == 1:  
15                 a[i][j] = 1  
16  
17 count = 0  
18  
19 for j in range(n):  
20     if j != source and a[source][j] == 1:  
21         count = count + 1  
22  
23 print(count)
```

```
1 INF = 999
2
3 a = [
4     [0, 2, INF],
5     [2, 0, 3],
6     [INF, 3, 0]
7 ]
8
9 n = 3
10
11 for k in range(n):
12     for i in range(n):
13         for j in range(n):
14             if a[i][k] + a[k][j] < a[i][j]:
15                 a[i][j] = a[i][k] + a[k][j]
16
17 diameter = 0
18
19 for i in range(n):
20     for j in range(n):
21         if a[i][j] != INF and a[i][j] > diameter:
22             diameter = a[i][j]
23
24 print(diameter)
```

```
1 INF = 999
2
3 a = [
4     [0, 3, INF, 7],
5     [8, 0, 2, INF],
6     [5, INF, 0, 1],
7     [2, INF, INF, 0]
8 ]
9
10 u = 0
11 v = 3
12 n = 4
13
14 for k in range(n):
15     for i in range(n):
16         for j in range(n):
17             if a[i][k] + a[k][j] < a[i][j]:
18                 a[i][j] = a[i][k] + a[k][j]
19
20 print(a[u][v])
```

```

2     [0, 1, 1],
3     [0, 0, 1],
4     [0, 0, 0]
5 ]
6
7 n = 3
8
9 for k in range(n):
10     for i in range(n):
11         for j in range(n):
12             if a[i][k] == 1 and a[k][j] == 1:
13                 a[i][j] = 1
14
15 answer = -1
16
17 for j in range(n):
18     possible = True
19
20     for i in range(n):
21         if i != j and a[i][j] == 0:
22             possible = False
23             break
24
25     if possible:
26         answer = j
27         break
28
29 print(answer)

```

```

3     [0, 0, 1, 0],
4     [0, 0, 0, 1],
5     [0, 0, 0, 0]
6 ]
7
8 n = 4
9
10 for k in range(n):
11     for i in range(n):
12         for j in range(n):
13             if a[i][k] == 1 and a[k][j] == 1:
14                 a[i][j] = 1
15
16 answer = 0
17 maximum = -1
18
19 for i in range(n):
20     count = 0
21
22     for j in range(n):
23         if i != j and a[i][j] == 1:
24             count = count + 1
25
26     if count > maximum:
27         maximum = count
28         answer = i
29
30 print(answer)

```



```
1 INF = 999
2
3 a = [
4     [0, 2, INF],
5     [2, 0, 4],
6     [INF, 4, 0]
7 ]
8
9 n = 3
10
11 for k in range(n):
12     for i in range(n):
13         for j in range(n):
14             if a[i][k] + a[k][j] < a[i][j]:
15                 a[i][j] = a[i][k] + a[k][j]
16
17 diameter = 0
18
19 for i in range(n):
20     for j in range(n):
21         if a[i][j] != INF and a[i][j] > diameter:
22             diameter = a[i][j]
23
24 print(diameter)
```



```
1 a = [  
2     [0, 1, 0],  
3     [0, 0, 1],  
4     [1, 0, 0]  
5 ]  
6  
7 n = 3  
8  
9 for k in range(n):  
10     for i in range(n):  
11         for j in range(n):  
12             if a[i][k] == 1 and a[k][j] == 1:  
13                 a[i][j] = 1  
14  
15 strong = True  
16  
17 for i in range(n):  
18     for j in range(n):  
19         if i != j and a[i][j] == 0:  
20             strong = False  
21             break  
22  
23 if strong:  
24     print("Strongly Connected")  
25 else:  
26     print("Not Strongly Connected")
```



Strongly Connected